

Day 19 - HyperParameter Tuning

Tuned candidates

Logistic Regression

- **C : Controls regularization.** [0.1,1,10]
Small C → More regularization.
→ Simpler model.
→ Less overfitting.
Large C → Less regularization.
→ More complex model.
- **Solver : optimization algorithm.**
lbfgs :
→ Modern optimizer.
→ Fast.
→ Works well for small/medium datasets.
liblinear :
→ Older optimizer.
→ Good for small datasets.

SVM

- **C : Controls regularization.** [0.1,1,10]
Small C → More regularization.
→ Simpler model.
→ Less overfitting.
Large C → Less regularization.
→ More complex model.
- **Kernel : Kernel determines the shape of the decision boundary.**
Linear Kernel → Straight line. Useful when data is linear
RBF Kernel → Curved boundary. Useful when data is nonlinear

Best

parameter
after applying
Grid search cv

{'C': 0.1, 'solver': 'lbfgs'}

{'C': 0.1, 'kernel': 'linear'}

Evaluation Metrics Using Tuned Models

Model	Accuracy	Precision	Recall	F1-Score	BER
Logistic Regression	0.742	0.69	0.617	0.651	0.258
SVM	0.758	0.846	0.468	0.603	0.242

Conclusion:

- Grid Search was performed on the shortlisted SVM and Logistic Regression models.
- The best SVM configuration was Linear Kernel with C=0.1.
- The best Logistic Regression configuration was C=0.1 with lbfgs solver.
- Cross-validation accuracy reached 82.81% for SVM and 79.25% for Logistic Regression.
- Final test-set performance remained similar to the baseline results, indicating that the original models were already close to optimal for the current dataset.
- SVM remained the strongest candidate and Logistic Regression remained the second-best candidate for further robustness analysis.